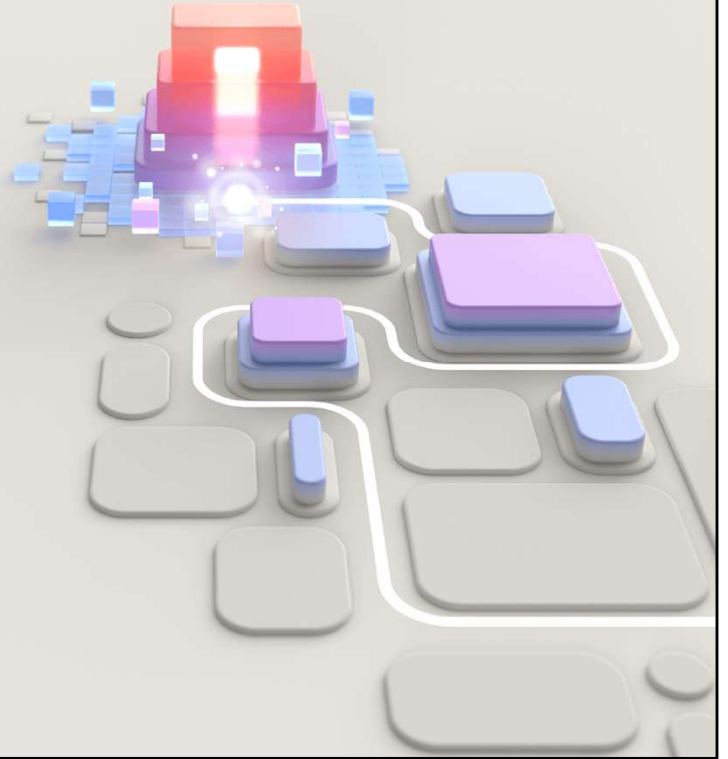




DP-300

Implement scalable database solutions using Azure SQL

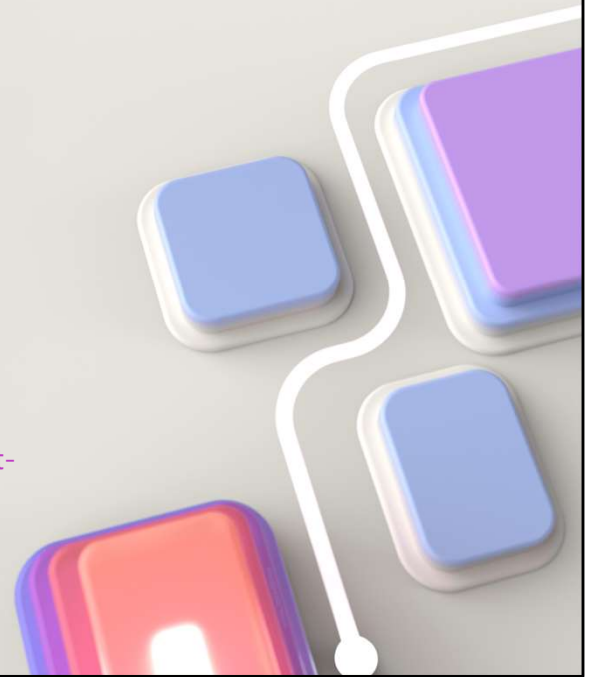


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Planning and implementing a high availability and disaster recovery environment

<https://learn.microsoft.com/training/paths/plan-implement-high-availability-disaster-recovery-environment/>

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<https://learn.microsoft.com/training/paths/plan-implement-high-availability-disaster-recovery-environment/>

Objectives

- The difference between recovery time and recovery point objectives
- The available HADR options
- The considerations for planning and configuring HADR solutions including how backup and restore fits in
- The factors that comprise a HADR strategy

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Data must be available when the business needs it. That means the solutions hosting the data must be designed with availability and recoverability in mind.

Suppose you work for a company that sells widgets both in stores and online. Your main application uses a highly transactional database for orders. What would happen if the server or platform hosting the transactional database had a problem that made it unavailable or inaccessible for some reason? What impact would it have on the business? If the right solution is put in place, the database would come online in a reasonable timeframe with not a lot of effort, thus allowing business to continue with minimal-to-no impact.

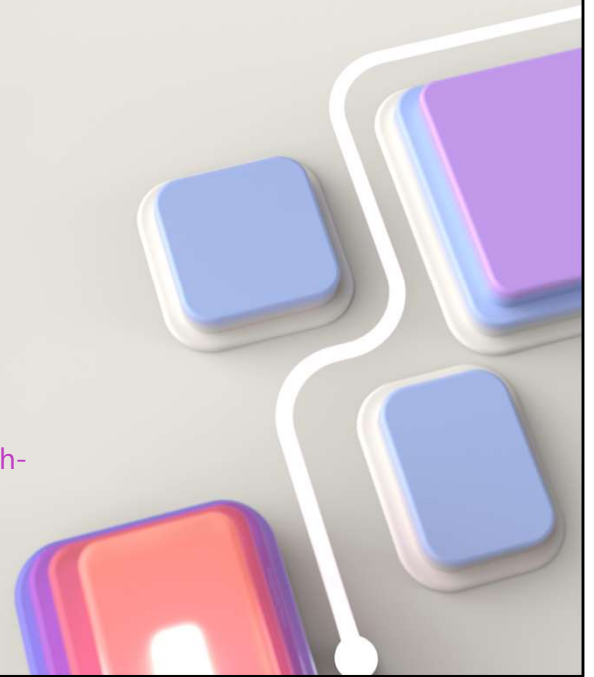
This Learning Path and its associated labs cover configuring, testing, and managing a solution for high availability and disaster recovery (HADR) in Azure for both Infrastructure-as-a-Service (IaaS) and Platform-as-a-Service (PaaS) deployments.

This Learning Path will not only cover basic requirements, but also the various options available to achieve HADR, , and will provide some hands-on experience with some of this through the lab.

Describe high availability and disaster recovery strategies

<https://learn.microsoft.com/training/modules/describe-high-availability-disaster-recovery-strategies/>

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Objectives



Recovery time objective (RTO)



Recovery point objective (RPO)



The available HADR options



How to devise a HADR strategy

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Implementing HADR database platform solutions is more than just deploying a feature. You must understand why and what you are doing to implement the right strategy.

Suppose you need to implement HADR for an IaaS VM that is running SQL Server 2019. Do you know how much time you would have to bring things back online if there was a problem? Do you know what options are available in SQL Server and the reasons for choosing one option over another?

By the end of this lesson, you will have a solid foundation that will allow you to implement HADR solutions in Azure.

Recovery Time Objective

Recovery Time Objective (RTO) is defined as the maximum amount of time to get a database or system up and running after an outage or problem

May have financial or business consequences if RTO is exceeded

When calculating RTO you must account for processes such as copying backup files, network speed

RTO can be defined for the entire system or for individual components

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RTO is the maximum amount of time technically allowed to bring resources online after an outage or problem. If that process takes longer than the RTO, there could be consequences such as financial penalties, work not able to be done, and so on. RTO can be defined for the whole solution which includes all aspects as well as individual components such as SQL Server instances and databases.

Recovery Point Objective

Recovery Point Objective (RPO) represents the maximum amount of data loss that is acceptable to the business

RPO is the point in time to which a database needs to be recovered

Zero data loss may be an option but often not a guarantee

Other environmental factors interfere with realizing ideal RPO

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RPO is the point in time to which a database needs to be recovered and defines the maximum amount of data loss that is acceptable to the business. For example, an IaaS VM containing SQL Server experiences an outage at 10:00 AM. The databases within the SQL Server instance installed have an RPO of 15 minutes. No matter what feature or technology is used to bring that instance and its databases back, the expectation is the data can be recovered to 9:45 AM or less to ensure minimal to no data loss.

Defining Recovery Time Objective and Recovery Point Objective

-  Must be defined per component (i.e., SQL Server) as well as the entire application architecture
-  The ability to recover from an outage is only as good as its weakest link – i.e., single points of failure
-  Getting everything online may take hours – RTOs and RPOs are usually separately
-  RTOs and RPOs should be defined for both HA and DR
-  DR generally involves a new data center, with different components – SQL Server is just one of them
-  HA is considered a more localized event

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All RTOs and RPOs should be formally documented and revised periodically or as needed.

RTO should be defined at a component level (e.g. SQL Server) as well as for the whole solution for the entire application architecture. The ability to recover from an outage is only as good as its weakest link. For example, if SQL Server and its databases can be brought online in five minutes but it takes application servers 20 minutes to do the same, the overall RTO would be 20 minutes, not five. The SQL Server environment could still have an RTO of five minutes; it still will not change the overall time to being live.

RPO is a data concern and directly influences the design of any HADR solution as well as administrative policies and procedures. The features used must support both the RTO and RPOs that are defined. For example, if transaction log backups are scheduled every 30 minutes but there is a 15-minute RPO, a database could only be recovered to the last transaction log backup available which in a best case would be 30 minutes ago. This assumes no other issues and the backups have been tested and are known to be good.

Infrastructure-as-a-Service vs. Platform-as-a-Service

IaaS solutions

- You use a combination of **platform availability options**, in conjunction with **database level solutions**
- In **IaaS**, there is **no built-in** high availability or disaster recovery at the database or the operating system level

PaaS solutions

- In **PaaS** solutions like Azure SQL Database, and Azure SQL Managed Instance, **high availability is built into the platform**
- In **PaaS and IaaS**, Disaster Recovery is an **optional** configuration

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IaaS solutions you use a combination of **platform availability options** like Availability Zones and Availability sets, in conjunction with **database level solutions** like Availability Groups or Log Shipping. The Azure platform does provide a level of availability, but a solution based on a single VM is subject to hardware failure, and downtime due to OS or RDBMS patching.

In **PaaS and IaaS**, Disaster Recovery (replication to another region) is an **optional** configuration

When it comes to availability, the choice of IaaS or PaaS makes a difference. With IaaS, you have a virtual machine (VM), which means there is an operating system with an installation of SQL Server. The administrator or group responsible for SQL Server would have a choice of HADR solutions and a great deal of control over what how that solution was configured.

With PaaS-based deployments such as Azure SQL Database, the HADR solutions are built into the feature and often just need to be enabled. There are minimal options that can be configured.

SQL Server HADR features for Azure virtual machines

- **Instance level protection** means that users and jobs are available on failover
- **Database level protection** means that everything inside of the user database is synchronized, but system objects in Master and MSDB require manual synchronization

Feature	Protects
Always On Failover Cluster Instance (FCI)	Instance
Always On Availability Group (AG)	Database
Log Shipping	Database

Both FCIs and AGs require an underlying cluster mechanism

For SQL Server deployments running on **Windows Server** use Windows Server Failover Cluster (WSFC), and for **Linux** use Pacemaker

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Both FCIs and AGs require an underlying cluster mechanism. For SQL Server deployments running on Windows Server, it is a Windows Server Failover Cluster (WSFC) and for Linux it is Pacemaker.

An instance of SQL Server is the entire installation of SQL Server (binaries, all the objects inside the instance including things like logins, SQL Server Agent jobs, and databases). Instance-level protection means that the entire instance is accounted for in the availability feature.

A database in SQL Server contains the data that end users and applications use. There are system databases that SQL Server relies as well as databases created for use by end users and applications. An instance of SQL Server always has its own system databases. Database-level protection means that anything that is captured in the transaction log for a user or application database is accounted for as part of the availability feature. Anything that exists outside of the database or that is not captured as part of the transaction log such as SQL Server Agent jobs and linked servers must be manually dealt with to ensure the destination server can function like the primary in the event of a planned or unplanned failover event.

Always On Failover Cluster Instance

Instance level protection

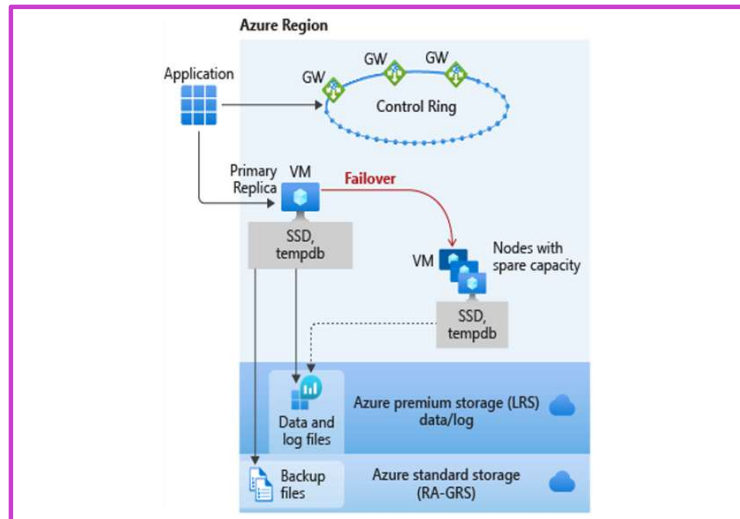
- Objects created at the instance level are protected

Failover process is a full stop and start of SQL Server

- Failover can be automatic or manual

Requires

- An underlying cluster
- A shared disk storage
- An Azure Load Balancer



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An FCI protects the entire instance. That means in the event of a problem, the entire instance would restart on another server known as a node. The failover process, whether automatic or manual, is a full stop and start of SQL Server which means it goes through recovery when starting on the other node. One big benefit of implementing an FCI is that everything in the instance (logins, users, jobs, linked servers, and so on) is available upon failover. There are no additional tasks to ensure it is ready for work in the event of the FCI switching hosts. For many, FCIs are the traditional form of SQL Server high availability that they implemented on premises.

FCIs only require one copy of a database, but that is also its single point of failure. To ensure another node can access the database, FCIs require some form of shared storage. For Windows Server-based architectures, this can be achieved via an Azure Premium File Share, iSCSI, Azure Shared Disk, Storage Spaces Direct (S2D), or a supported third-party solution like SIOS DataKeeper. FCIs using Standard Edition of SQL Server can have up to two nodes. FCIs also require the use of Active Directory Domain Services (AD DS) and Domain Name Services (DNS), so that means AD DS and DNS must be implemented somewhere in Azure for it to work.

- Requires an underlying cluster (Windows Server Failover Cluster or Pacemaker)
- Requires a shared disk storage
- Clients connect to FCI virtual network name or IP address, not the node
- Standard Edition of SQL Server can have up to two nodes
- Requires an Azure Load Balancer

Always On Availability Group

- Provides high availability, and requires an underlying cluster
- There is a read-scale option, but that model does not support failover
- Availability groups have **no shared storage requirement**, each replica maintains a full copy of each database in the group

Standard Edition

Supports only two replicas (one primary and one secondary), one database per AG

Enterprise Edition

Supports up to nine replicas per AG (one primary and up to 8 secondaries), and can have multiple databases in an AG

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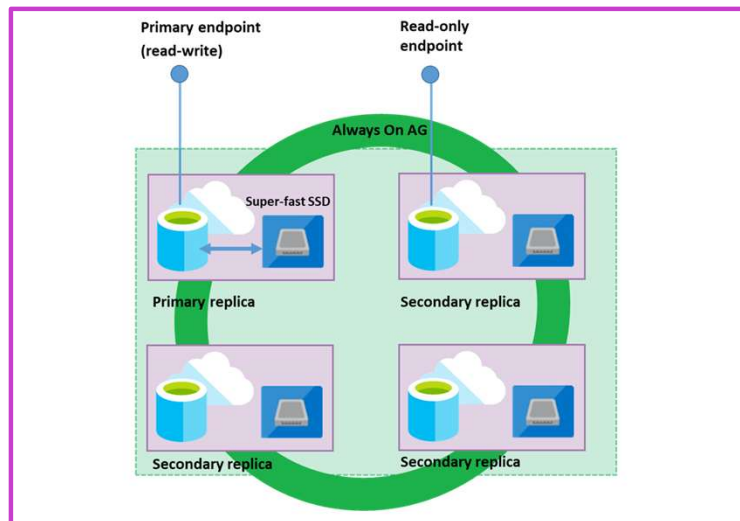
An AG provides database-level protection. In Standard Edition, an AG can contain one database whereas in Enterprise Edition, an AG can have more than one database. There is one fully read-write copy of the database in an AG which exists on the primary replica. The other instances participating in the AG are secondary replicas where the database is in a loading state. As transactions occur on the primary replica, they are sent to the secondary replica(s) either synchronously or asynchronously. Unless a secondary replica is lagging for some performance reason, the failover process is not only quick, but there could be minimal to no data loss on a failover.

A secondary replica can be used for more than just HADR. It can be configured for generating backups, running DBCC commands, and read-only queries. All of these are Enterprise Edition-only features.

The AG Listener provides a virtual network name and IP address

Always On Availability Group

- Can be used in **IaaS** or **PaaS**
- Database level protection
- The primary replica is read/write and the secondary replica is Read Only
- Replication can be **synchronous** or **asynchronous**.
- Use Microsoft Entra authentication or a contained database



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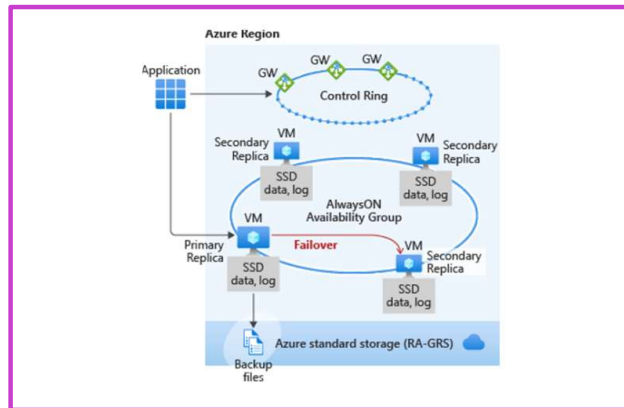
- Requires an ILB.
- Any object outside of the database or not captured in the transaction log, must manually be created on the new primary (SQL Server Agent jobs, instance-level logins, and linked servers).

An FCI protects the entire instance. That means in the event of a problem, the entire instance would restart on another server known as a node. The failover process, whether automatic or manual, is a full stop and start of SQL Server which means it goes through recovery when starting on the other node. One big benefit of implementing an FCI is that everything in the instance (logins, users, jobs, linked servers, and so on) is available upon failover. There are no additional tasks to ensure it is ready for work in the event of the FCI switching hosts. For many, FCIs are the traditional form of SQL Server high availability that they implemented on premises.

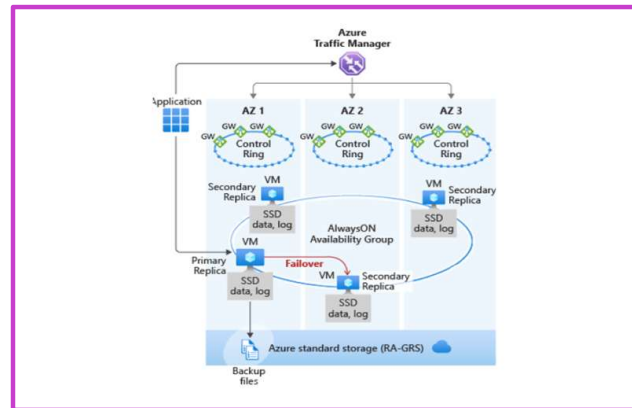
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Always On Availability Group

One region



Multiple zones



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Den här bilden visar hur **SQL Server Always On Availability Groups** kan byggas i Azure för hög tillgänglighet. Till vänster ser du en lösning inom **en och samma region**, där en primär replik och en eller flera sekundära repliker körs på separata VM:ar. Om den primära noden går ner sker en **failover**, så att en sekundär replik tar över och applikationen kan fortsätta arbeta. I samma region kan detta ge mycket bra tillgänglighet och relativt låg fördröjning mellan replikerna.

Till höger visas en mer robust variant med **flera Availability Zones (AZ1, AZ2, AZ3)**. Här sprids replikerna mellan olika zoner i samma region, vilket skyddar mot zonfel, t.ex. ström- eller nätverksproblem i ett datacenter. Ofta används en lastbalanserings-/routing-komponent (i bilden Azure Traffic Manager) för att styra anslutningar till rätt instans. Båda varianterna kompletteras med **backup till Azure Storage (RA-GRS)** för extra skydd. Sammanfattningsvis: en region ger HA, flera zoner ger HA + bättre skydd mot större infrastrukturfel.

The underlying database files (.mdf/.ldf) are placed on the attached SSD storage to provide very low latency IO to your workload. High availability is implemented using a technology similar to SQL Server [Always On availability groups](#). The cluster includes a single primary replica that is accessible for read-write customer workloads, and up to three secondary replicas (compute and storage) containing copies of data. The primary node constantly pushes changes to the secondary nodes in order and ensures that the data is persisted to at least one secondary replica before committing each transaction. This process guarantees that if the primary node crashes for any reason, there is always a fully synchronized node to fail over to. The failover is initiated by the Azure Service Fabric. Once the secondary replica becomes the new primary node, another secondary replica is created to ensure the cluster has enough nodes (quorum set). Once failover is complete, Azure SQL connections are automatically redirected to the new primary node.

Because the zone-redundant databases have replicas in different datacenters with some distance between them, the increased network latency may increase the commit time and thus impact the performance of some OLTP workloads. You can always return to the single-zone configuration by disabling the zone-redundancy setting. This process is an online operation similar to the regular service tier upgrade. At the end of the process, the database or pool is migrated from a zone-redundant ring to a single zone ring or

vice versa.

Log shipping

- Based on backup and restore to provide user database level protection. Doesn't include SQL Server Agent jobs, linked servers, instance-level logins, etc.
- Automated process for backing up, copying, and applying transaction log backups to provide the **simplest method** of achieving **HADR** for SQL Server. **Primarily used for DR**, but it could also be used for **HA**.
- Transaction log loads always slightly delayed.
- Protects against human error (accidental deletion of data).

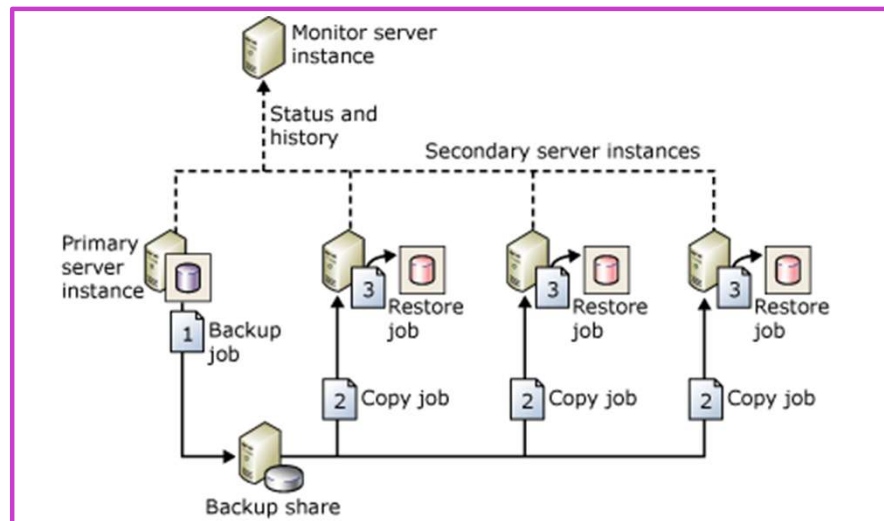
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Log shipping has been around since the early days of SQL Server. It is based on backup, copy, and restore and is one of the simplest methods of achieving HADR for SQL Server. Log shipping is primarily used by most for disaster recovery, but it could be used to enhance local availability, too.

With log shipping, you take a full backup of a database, restore it in a loading state on another instance, and then SQL Server will automatically backup, copy, and restore transaction log backups.

Log shipping

- When switching to secondary, application must be able to tolerate a name change or use a DNS alias
- Can configure automatic replication on a **schedule** to **backup, copy and restore** the primary database's transaction log to secondary

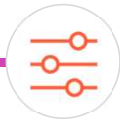


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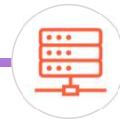
Azure HADR features for Azure Virtual Machines



**Availability
sets**



**Availability
zones**



**Azure site
recovery**

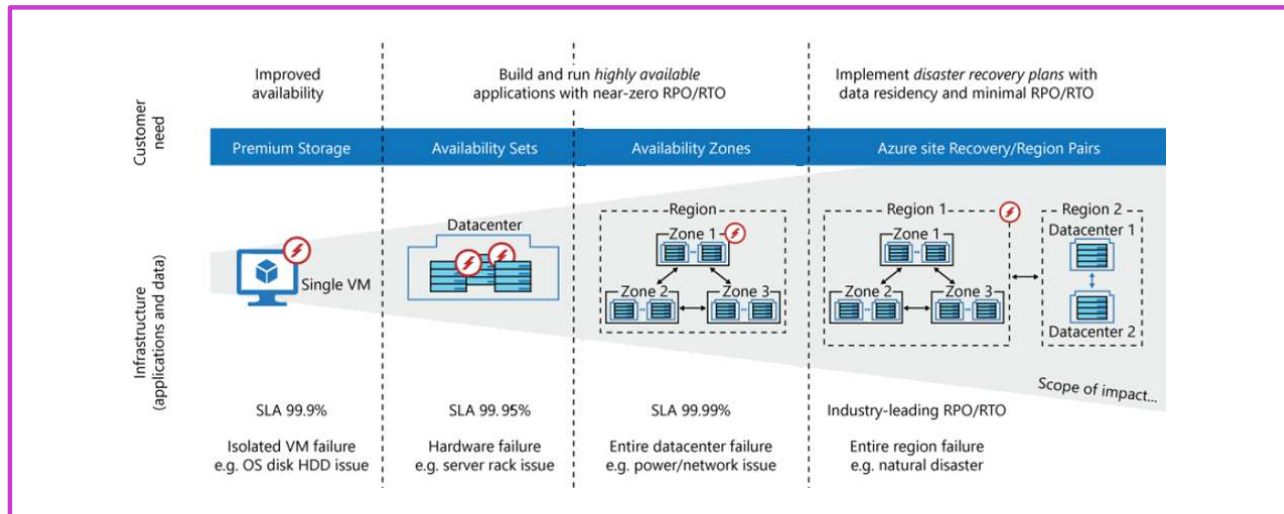
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Azure provides three main options to enhance availability for IaaS deployments:

- Availability Sets
- Availability Zones
- Azure Site Recovery

All three of these options are external to the VM and do not know what kind of workload is running inside of it.

High Availability offered by a single VM, Availability Sets, and Availability Zones, and Azure Site Recovery



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Den här bilden jämför olika sätt att få **hög tillgänglighet (High Availability)** i Azure – från enklast till mest robust. Längst till vänster ser vi en **Single VM**. Den kan ha bra prestanda och t.ex. Premium Storage, men om själva VM:en får problem (t.ex. OS-diskfel) påverkas hela tjänsten. SLA ligger runt **99,9%**.

Nästa steg är **Availability Sets**, där två eller fler VM:ar placeras i samma datacenter men i olika "fault domains" och "update domains". Då skyddas du mot t.ex. hårdvarufel i en rack eller planerade uppdateringar. SLA förbättras till ca **99,95%**.

Sedan kommer **Availability Zones**, där resurserna placeras i separata zoner (olika datacenter) inom samma region. Det skyddar mot större fel som kan slå ut ett helt datacenter, exempelvis ström- eller nätverksproblem. Här kan SLA vara runt **99,99%**.

Sist visas **Azure Site Recovery / Region Pairs**, vilket är mer av en katastrofåterställningslösning (Disaster Recovery). Då kan du replikera till en annan region och överleva ett regionalt avbrott, t.ex. naturkatastrof. Denna nivå handlar mest om RPO/RTO: hur mycket data du kan förlora och hur snabbt du kan återstarta.

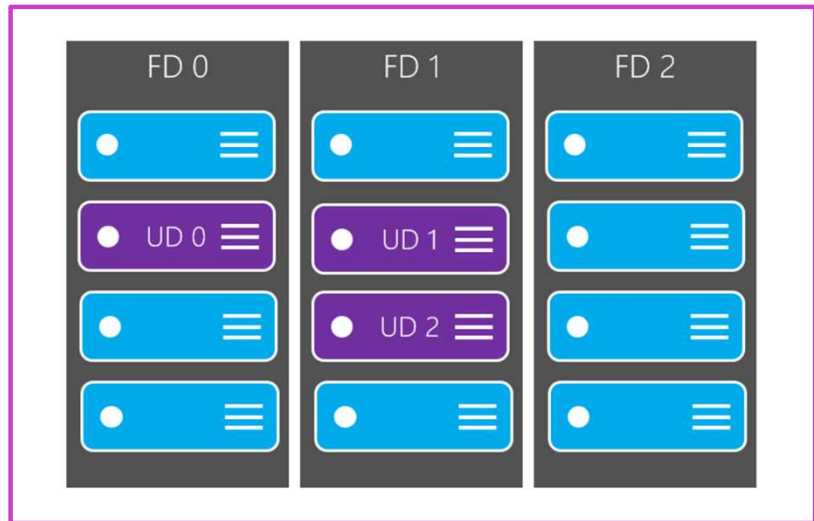
Azure provides three main options to enhance availability for IaaS deployments:

- Availability Sets
- Availability Zones
- Azure Site Recovery

All three of these options are external to the VM and do not know what kind of workload is running inside of it.

Availability sets

- Protects against physical hardware failure, network outages, power interruptions and patching update outages
- **Fault domain:**
 - Up to 3
 - VM will not share the same power source, network switch
- **Update domain:**
 - Up to 5
 - Controls reboot behavior
- **Does not protect against a failure in the VM's OS or RDBMS**



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Availability Sets används i Azure för att öka tillgängligheten för virtuella maskiner genom att sprida dem över olika fysiska resurser. De skyddar mot hårdvarufel, nätverksavbrott, strömproblem och planerade uppdateringar. Detta görs med två begrepp: **Fault Domains (FD)** och **Update Domains (UD)**. Fault domains (upp till 3) innebär att VM:ar inte delar samma strömkälla eller nätverksswitch, så ett rackfel slår inte ut allt. Update domains (upp till 5) styr patchning och omstarter så att inte alla VM:ar uppdateras samtidigt. Availability Sets skyddar dock inte mot fel i operativsystemet eller SQL Server själv.

Availability Sets provide uptime against Azure related maintenance and single points of failure in a single data center. This was one of the first availability features introduced into the Azure platform, and effectively it can be thought of as anti-affinity rules for your VMs. This means if you had two SQL Server VMs in an availability set or log shipping pair, they would be guaranteed to never run on the same physical server.

Availability Zones

- VMs (or other Azure resources) are placed in separate datacenters in the same Azure region
- Enables survival of a datacenter
- Does not protect against a failure in the guest
- Latency should be low enough to support synchronous data transfer between zones

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Availability Zones är ett sätt att bygga högre tillgänglighet i Azure genom att placera virtuella maskiner (och andra resurser) i **separata datacenter** inom **samma Azure-region**. Varje zon är en egen fysisk plats med egen strömförsörjning, kylning och nätverk. Poängen är att om ett helt datacenter får problem – exempelvis strömbavbrott eller större hårdvarufel – kan tjänsten fortsätta köras i de andra zonerna. Det gör

Availability Zones starkare än Availability Sets, eftersom de klarar bortfall av ett helt datacenter.

Samtidigt skyddar Availability Zones **inte** mot fel inne i själva VM:en, t.ex. kraschat operativsystem, felaktiga drivrutiner eller problem i SQL Server-processen. För det krävs fortfarande redundans på applikations-/databassidan, t.ex. Always On Availability Groups eller annan replikeringslösning.

En viktig förutsättning är att **latensen mellan zonerna är låg**, så att synkron dataöverföring fungerar bra. Det gör att man kan använda synkron replikering mellan zoner utan att prestandan blir orimligt lidande, och samtidigt få mycket bra skydd mot zon-/datacenterfel.

Availability zones build on the infrastructure in Availability Sets, but provide your application with availability in the event of a data center failure in Azure. Each Azure region consists of many data center with low latency network connections between them. When you deploy VM resources in a region that supports Availability Zones, you have the option to deploy those resources into Zone 1, 2, or 3.

You should note that the zone numbers in the portal are logical representations. So that if you and I both deploy a VM into Zone 1 in our own subscriptions, that does not mean those VMs exist in the same physical Azure Data Center. Additionally, because of the distance there can be some additional latency introduced into zonal deployments. You should test the latency between your VMs to ensure that the latency meets the application's performance targets. In most cases round trip latency will be less than 1 millisecond, which supports synchronous replication in features like availability groups. You can also deploy Azure SQL Database into Availability Zones.

Azure Site Recovery

- **SQL Server agnostic** – knows nothing of SQL Server transactions
- Supports hybrid disaster recovery
- Replicates at the VM disk level; VMs might need to be grouped together
- Frequently used for migration from on-premises to Azure
- Can test DR without bringing down production

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Azure Site Recovery (ASR) är en tjänst för **disaster recovery (DR)** som replikerar hela virtuella maskiner mellan platser, t.ex. från on-premises till Azure, eller mellan regioner i Azure. Den stora poängen är att ASR är **SQL Server-agnostisk**: den bryr sig inte om databaser, transaktioner eller loggkedjor – den replikerar i stället på **VM/disk-nivå**. Det gör ASR väldigt flexibel, men också att den inte kan garantera samma databaskonsistens som SQL-specifika tekniker (t.ex. Always On Availability Groups).

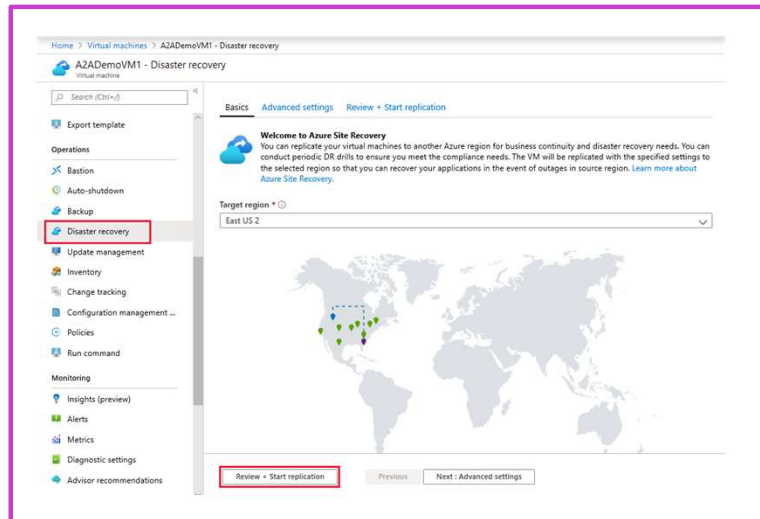
ASR används ofta i **hybrid-scenarier**, där man vill ha en sekundär miljö i Azure redo om den lokala miljön fallerar. Eftersom replikeringsenheten är VM:en kan man också behöva **gruppera flera VM:ar** som hör ihop (t.ex. webbserver + applikationsserver + SQL-server) så att de startas i rätt ordning vid failover.

En annan vanlig användning är **migrering**: man replikerar VM:ar till Azure och gör sedan en kontrollerad failover som "cutover". En stor fördel är att du kan **testa DR utan att stoppa produktion**, genom att göra test-failover till ett isolerat nätverk. Det gör ASR till ett praktiskt verktyg både för kontinuitetsplanering och förflyttning till molnet.

ASR is an option that deals with the VM but is agnostic to the fact that SQL Server is running inside of it. It works with SQL Server but is not designed specifically account for nuances that may be required when you have a specific RPO. The disks are replicated to another region. This means that all changes to a disk are sent as soon as they occur, but this process knows nothing of database transactions. This is why recovering to a specific data point may not be possible with ASR in the same way it is for something like an AG

Azure Site Recovery – DR example

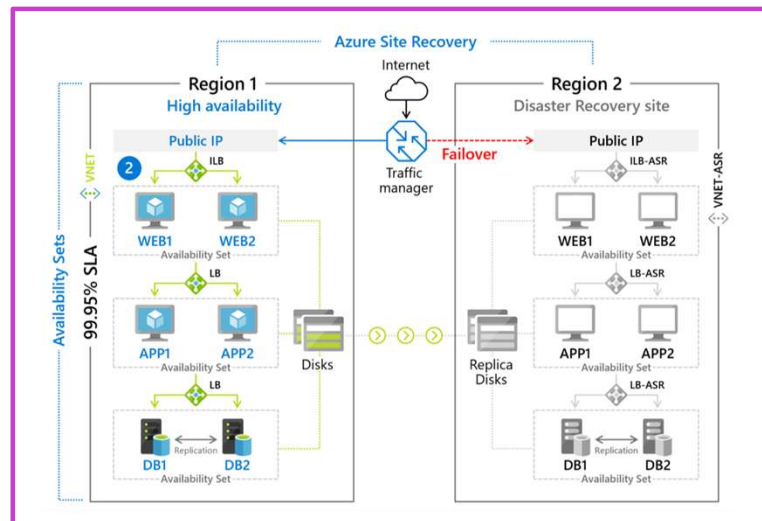
- Not a SQL Server-based disaster solution
- Most prefer a **database-centric** approach to lower RPO
- ASR can be used with more than just SQL Server
- ASR may meet RTO and RPO
- ASR is provided as part of the Azure platform



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Azure Site Recovery

- Replicate Azure VMs from one Azure region to another
- Replicate on-premises VMware VMs, Hyper-V VMs, physical servers (Windows and Linux), Azure Stack VMs to Azure
- Replicate AWS Windows instances to Azure



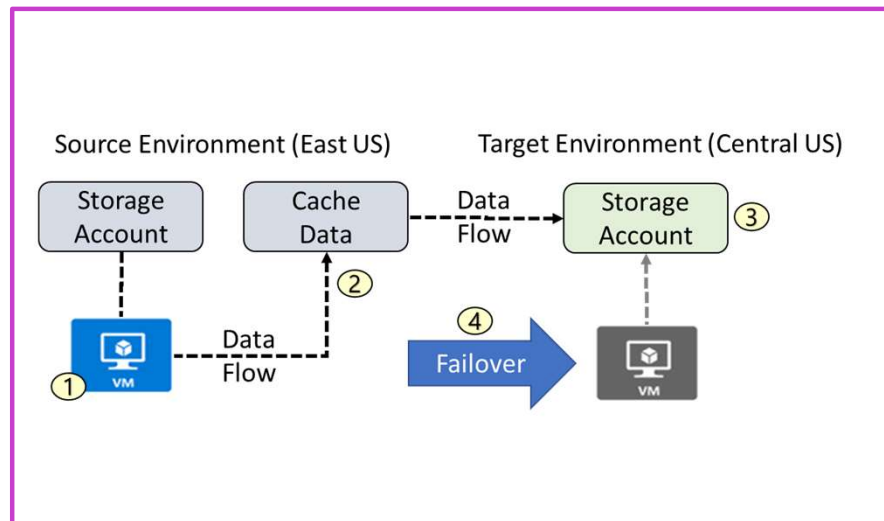
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Azure Site Recovery (ASR) är en DR-tjänst som **replikerar hela maskiner** till en sekundär plats så att du snabbt kan göra **failover** vid större avbrott. Den kan kopiera **Azure-VM:ar mellan regioner**, men också flytta arbetslaster från **on-premises** (VMware, Hyper-V eller fysiska servrar för både Windows och Linux) till Azure. Även **Azure Stack-VM:ar** och till och med **AWS Windows-instanser** kan replikeras till Azure. Vid en incident startas dessa repliker i Region 2, och trafik kan styras om (t.ex. via Traffic Manager) så att användarna fortsätter nå tjänsten.

As a fully integrated offering, Site Recovery is automatically updated with new Azure features as they're released. Minimize recovery issues by sequencing the order of multi-tier applications running on multiple virtual machines. Ensure compliance by testing your disaster recovery plan without impacting production workloads or end users. And keep applications available during outages with automatic recovery from on-premises to Azure or Azure to another Azure region.

Azure Site Recovery cont'd

1. VM is registered with Azure Site Recovery
2. Data is continuously replicated to cache
3. Cache is replicated to the target storage account
4. During failover the virtual machine is added to the target environment



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Azure Site Recovery (forts.) visar hur replikeringen går till steg för steg mellan två regioner. Först **registreras VM:en** i Azure Site Recovery. Därefter **replikeras data kontinuerligt** från den körande VM:en till en **cache** i källmiljön (t.ex. East US). Cachen fungerar som en mellanlagring som samlar upp förändringar innan de skickas vidare. Sedan **replikeras cache-data** över till ett **Storage Account i målmiljön** (t.ex. Central US), vilket gör att en uppdaterad kopia av VM:ens diskdata finns redo i andra regionen. När ett avbrott inträffar initieras **failover**, och då skapas/aktiveras VM:en i målregionen med hjälp av den replikerade datan.

På så sätt kan tjänsten snabbt startas upp i en annan region utan att man först behöver återställa från backup.

As a fully integrated offering, Site Recovery is automatically updated with new Azure features as they're released. Minimize recovery issues by sequencing the order of multi-tier applications running on multiple virtual machines. Ensure compliance by testing your disaster recovery plan without impacting production workloads or end users. And keep applications available during outages with automatic recovery from on-premises to Azure or Azure to another Azure region.

Hybrid solutions

On premises (physical or virtual) with an Azure component:

Includes backing up to Azure as cold storage

Can go Azure to or from another cloud provider or on-premises

Networking is the most important component to consider:

- ExpressRoute from on premises to Azure
- Site to Site VPN for everything else
- Need adequate bandwidth
- Must be secure

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Den här sliden sammanfattar vad **hybridlösningar** innebär i Azure-sammanhang: att man kombinerar **on-premises** (fysiska eller virtuella servrar i eget datacenter) med **Azure-komponenter**. Ett vanligt scenario är att behålla system lokalt men använda Azure för exempelvis **backup** eller billigare **cold storage**. Hybrid kan också betyda att man kopplar ihop Azure med **andra molnleverantörer** eller flyttar arbetslasten mellan miljöer. Den viktigaste faktorn är **nätverk**, eftersom det avgör prestanda, stabilitet och säkerhet. För hög kvalitet och låg latens används ofta **ExpressRoute** mellan on-premises och Azure. För enklare eller billigare kopplingar används **site-to-site VPN**. Oavsett krävs tillräcklig bandbredd och stark säkerhet.

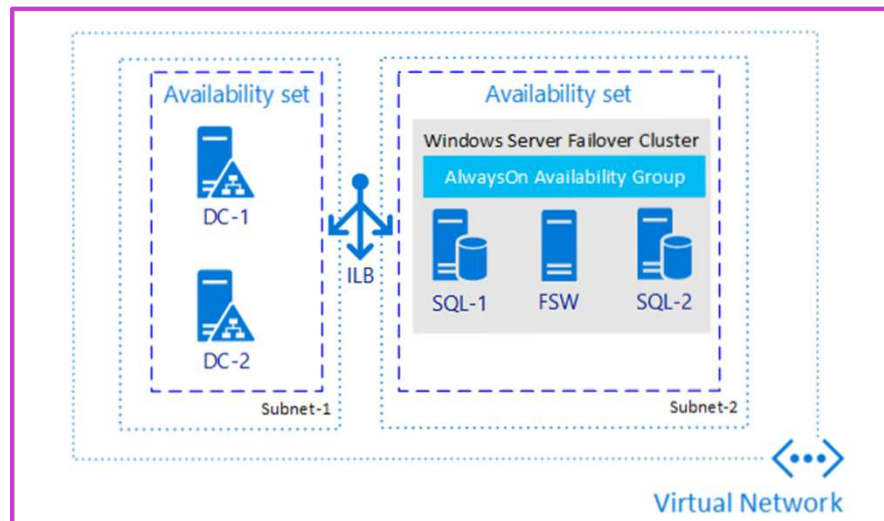
While an architecture can be completely deployed in one or more Azure regions, many will need or want to have solutions that span both on premises and Azure, or possibly Azure to another public cloud. This is known as a hybrid.

PaaS solutions by nature are not designed to allow traditional hybrid solutions. HADR is provided by the Azure infrastructure. There are some exceptions. For example, SQL Server's transactional replication feature can be configured from on premises (or another cloud) to Azure SQL Database Managed Instance, but not the other way.

Hybrid solutions are therefore IaaS-based since they rely on traditional infrastructure. Hybrid solutions are extremely useful. Not only can they be used to help migrate to Azure, but the most common scenario is to create a robust disaster recovery solution for on premises. For example, a secondary replica for an AG can be added in Azure. That means any associated infrastructure must exist such as AD DS and DNS.

HA Single Region Example – Availability Group

- Protects data by having more than one copy on different VMs
- No shared storage, so easier to set up than an FCI
- Provides enhanced availability during patching scenarios
- Meet RTO and RPO with minimal-to-no data loss
- ILB=Internal Load Balancer



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Den här bilden visar ett **High Availability (HA)**-exempel för **SQL Server Always On Availability Group** i **en och samma Azure-region**. Lösningen bygger på att databasen finns i **flera kopior på olika virtuella maskiner** (SQL-1 och SQL-2). På så sätt kan systemet fortsätta fungera även om en server får problem eller behöver startas om.

Till skillnad från en klassisk **Failover Cluster Instance (FCI)** kräver Availability Groups **ingen delad disk/lagring** mellan noderna. Varje SQL-server har sin egen lagring, och data synkroniseras via replikering. Det gör ofta lösningen **enklare att sätta upp i Azure**, där delad lagring kan vara mer komplex.

Bilden visar också en **Windows Server Failover Cluster (WSFC)** som ger klusterfunktionaliteten (failover, quorum). För quorum används en **FSW (File Share Witness)**, som hjälper klustret att avgöra vilken nod som ska vara aktiv om något händer.

ILB (Internal Load Balancer) är central: den ger en **gemensam intern IP-adress** för klienter (t.ex. AG Listener). Klienterna ansluter alltid till samma adress, och ILB pekar automatiskt mot den nod som är **primär** efter en failover.

Availability Sets används för att sprida VM:arna över olika fel-/uppdateringsdomäner och därmed minska risken för samtidig nedtid vid underhåll.

Why is this architecture worth considering?

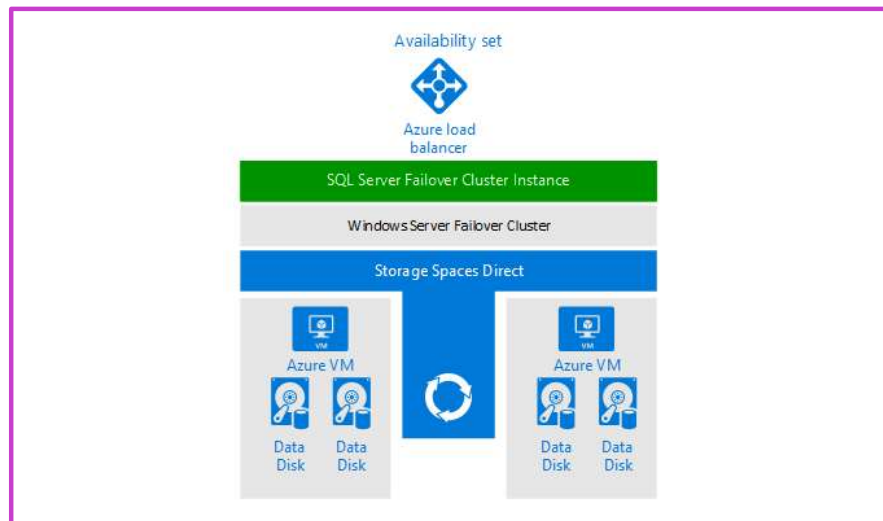
- Protects data by having more than one copy on different VMs
- Meet RTO and RPO with minimal-to-no data loss if implemented properly
- Provides an easy, standardized method for applications to access both primary and secondary replicas (if things like read only replicas will be used)

- Provides enhanced availability during patching scenarios
- No shared storage, so less complication than an FCI

Details on AGs in next Lesson

HA Single Region Example – Failover Cluster Instance

- Well established, proven solution
- Meets most RTO and RPO
- Provides enhanced availability during patching scenarios
- Designed to protect against network card failure or disk failure



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Den här bilden visar ett **HA-exempel i en och samma region** med **SQL Server Failover Cluster Instance (FCI)** i Azure. Det är den klassiska "kluster-SQL"-modellen där **en instans flyttas mellan noder** vid fel, istället för att flera SQL-servrar replikerar data (som i Availability Groups). Lösningen bygger på **Windows Server Failover Cluster (WSFC)** och använder **delad lagring** – i bilden representerat av **Storage Spaces Direct** – så att båda noderna kan komma åt samma databaser och loggar. Vid failover tar den andra VM:n över, och applikationen fortsätter ansluta via en **Azure Load Balancer** (ofta ILB) som ger en stabil anslutningspunkt.

Why is this architecture worth considering?

Well known, proven solution

Meets most RTO and RPO

Provides an easy, standardized method for applications to access the clustered instance of SQL Server

Provides enhanced availability during patching scenarios

I praktiken betyder detta att FCI är **väletablerat och beprövat**, och passar när man vill ha en tydlig och robust HA-lösning med snabb återstart (RTO) och låg risk för dataförlust (RPO). Eftersom applikationer ansluter till **en gemensam klusteradress** blir det dessutom ett standardiserat sätt att nå SQL Server, även när den aktiva noden byts vid underhåll eller incidenter.

Why is this architecture worth considering?

- Well known, proven solution
- Meets most RTO and RPO
- Provides an easy, standardized method for applications to access the clustered instance of SQL Server
- Provides enhanced availability during patching scenarios

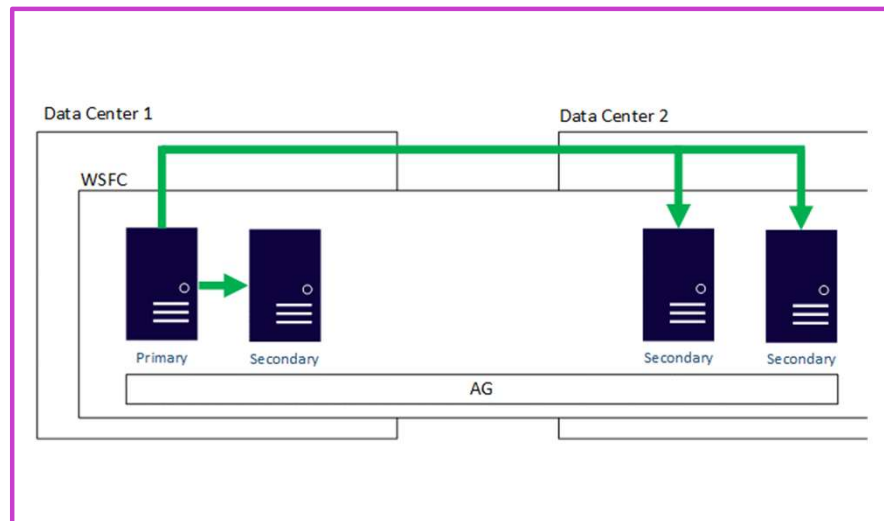
Availability Groups vs. Failover Cluster Instance

	Nodes within an FCI	Replicas within an availability group
Uses WSFC	Yes	Yes
Protection level	Instance	Database
Storage type	Shared	Non-shared
Storage solutions	Direct attached, SAN, mount points, SMB	Depends on node type
Readable secondaries	No*	Yes
Applicable failover policy settings	WSFC quorum FCI-specific Availability group settings**	WSFC quorum Availability group settings
Failed-over resources	Server, instance, and database	Database only

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DR example – Multi-region/Hybrid Availability Group

- Use with **Standard** and **Enterprise** editions of SQL Server
- Provides both HA and DR
- **All nodes** which contain the replicas participate in the same WSFC (assumes good network connectivity)
- Consider using a cloud witness for WSFC
- Requires AD DS and DNS



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Den här bilden visar ett **DR-exempel (Disaster Recovery)** med en **multi-region/hybrid Always On Availability Group**. Poängen är att du inte bara får hög tillgänglighet (HA) inom ett datacenter, utan även ett skydd mot större avbrott genom att ha **replicas i flera datacenter/regioner**. I figuren finns en **Primary** i Data Center 1 som skickar transaktionsloggar (gröna pilar) till en eller flera **Secondary**-replicas, både lokalt och i Data Center 2. Det gör att du kan hantera allt från serverfel till ett helt datacenter-bortfall.

Lösningen kräver **SQL Server Standard eller Enterprise**, och bygger på att alla noder ingår i samma **WSFC (Windows Server Failover Cluster)**. Det innebär att nätverket mellan platserna måste vara stabilt och ha bra latency, annars blir failover och synkronisering problematiskt. För att klustret ska fungera korrekt behövs även **AD DS och DNS**, eftersom Windows-kluster och SQL Server AG använder dessa för identitet, namnuppslagning och resurshantering.

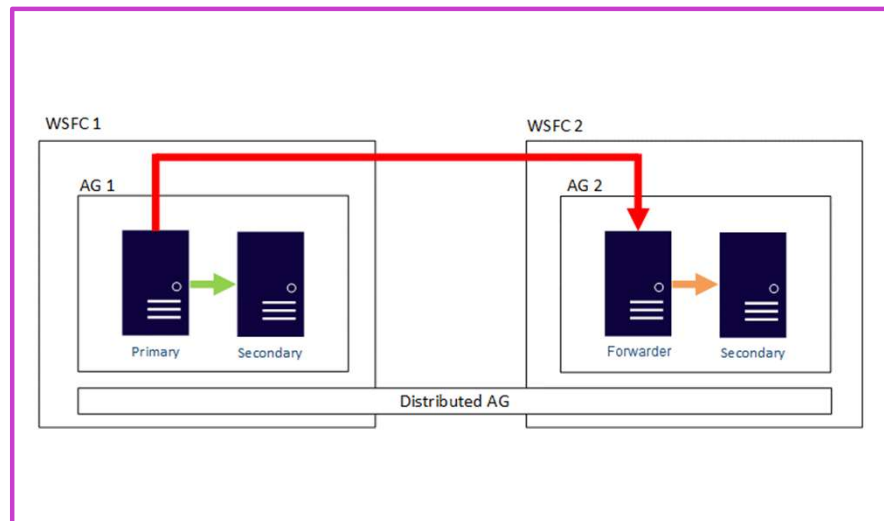
Eftersom man ofta vill undvika en "split brain"-situation vid nätverksproblem rekommenderas att använda en **cloud witness** (t.ex. i Azure Storage) som extra röst i klustrets quorum. Sammantaget är detta en kraftfull design som ger både **HA + DR**, men den kräver mer infrastruktur, nätverksplanering och korrekt konfiguration än en enkel AG i en enda region.

I

If you are using AGs, one option is to configure it across multiple Azure regions or potentially as a hybrid architecture. This means that all nodes which contain the replicas participate in the same WSFC. This assumes good network connectivity especially if this is a hybrid configuration, and one of the biggest considerations would be the witness resource for the WSC which will be discussed in Lesson 2. As noted above, this would require AD DS and DNS everywhere.

DR example – Distributed Availability Group

- Uses multiple WSFCs and multiple AGs
- Two WSFCs, no single point of failure
- Primary replica contains the read write database (global primary) and is **not** synchronizing all secondary replicas
- Used in **Azure** or **hybrid**, provides fail back from one location to another



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Den här bilden visar ett **DR-exempel med Distributed Availability Group (Distributed AG)** i SQL Server. Skillnaden mot en "vanlig" Availability Group är att man här kopplar ihop **flera separata WSFC-kuster och flera AG:er**. I figuren finns **WSFC 1 med AG 1** (Primary + Secondary) och **WSFC 2 med AG 2** (Forwarder + Secondary). Den röda pilen illustrerar att data replikeras mellan AG:erna via en **Distributed AG-koppling**.

Poängen är att du får en lösning för **multi-site/multi-region** där varje site kan ha sin egen lokala AG och sin egen klusterinfrastruktur. Det innebär **ingen single point of failure** på klusternivå – om ett helt kluster eller en region går ner kan du fortfarande failover till den andra platsen.

En viktig detalj är att **global primary** (read/write-databasen) inte synkar direkt till alla sekundärer överallt. Istället fungerar **Forwarder** som en mellanhand: den tar emot replikerad data från första AG:n och skickar den vidare till sin lokala secondary. Det gör modellen mer skalbar och bättre lämpad för **Azure eller hybrid-scenarier**, där man vill kunna flytta drift mellan platser och även stödja **failback** (tillbaka-failover) när primär plats är återställd.

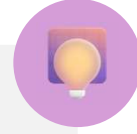
- **Enterprise** edition feature of SQL Server.
- The primary of the second AG is known as a forwarder and keeps the secondary replica(s) of that AG in sync.

A distributed AG is an Enterprise Edition feature introduced in SQL Server 2016. It is different than a traditional AG. Instead of having one underlying WSFC where all of nodes contain replicas as shown in Example 1, a distributed AG is made up of multiple AGs. The primary replica containing the read write database is known as the global primary. The primary of the second AG is known as a forwarder and keeps the secondary replica(s) of that AG in sync. In essence, this is an AG of AGs.

This architecture makes it easier to deal with things like quorum (discussed in more depth in Lesson 2) since each cluster would maintain its own. A distributed AG would work whether everything is all in Azure or this is a hybrid architecture.

More later on distributed AGs.

Knowledge check



What is RPO?

- ☐ The number of nodes in a cluster
- ☒ The point to which data needs to be recovered after a failure
- ☐ A partial database restore



What is a hybrid solution?

- ☒ A solution that has resources both in Azure as well as on premises or in another cloud provider
- ☐ A solution that uses two different database engines e.g. MySQL and SQL Server
- ☐ A solution that spans two different versions of SQL Server



What is available after failover with database-level protection in SQL Server?

- ☐ Logins, Databases, and SQL Server Agent Job
- ☐ Databases and SQL Server Agent jobs
- ☒ Whatever is in the databases; anything outside must be dealt with manually

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1) What is RPO?

☒ **The point to which data needs to be recovered after a failure**
RPO anger hur mycket data du kan förlora vid avbrott.

☒ **The number of nodes in a cluster**
Antal noder beskriver klusterstorlek, inte återställningspunkt.

☒ **A partial database restore**
Delåterställning är en metod, inte definitionen av RPO.

2) What is a hybrid solution?

☒ **A solution that has resources both in Azure as well as on premises or in another cloud provider**
Hybrid betyder blandning av Azure och lokal miljö/annan molnleverantör.

☒ **A solution that uses two different database engines e.g. MySQL and SQL Server**
Två databasmotorer är inte hybrid, bara teknisk variation.

☒ **A solution that spans two different versions of SQL Server**
Olika versioner är uppgradering/migrering, inte hybrid arkitektur.

3) What is available after failover with database-level protection in SQL Server?

☒ **Whatever is in the databases; anything outside must be dealt with manually**
Endast databasinnehåll skyddas; logins, jobb kräver extra hantering.

☒ **Logins, Databases, and SQL Server Agent Job**
Logins och Agent-jobb flyttas inte automatiskt vid failover.

☒ **Databases and SQL Server Agent jobs**
Agent-jobb ingår ej i databasskydd; måste synkas separat.

Explore IaaS and PaaS solutions for high availability and disaster recovery

<https://learn.microsoft.com/training/modules/explore-iaas-paas-platform-tools-for-high-availability-disaster-recovery/>

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If you are using IaaS for deploying your database solution, there are considerations for properly deploying HADR.

Suppose you want to use an AG. How is deploying a WSFC different? Are there any specific considerations for AGs that differ from an on-premises solution?

By the end of this lesson, you will understand what is needed to be able to deploy IaaS database platform solutions in Azure.

Objectives



What to consider when deploying a WSFC in Azure



What to consider when deploying an AG in Azure



Understand active-geo replication



Explore auto-failover groups

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If you're using IaaS for deploying your database solution, there are considerations for properly deploying HADR.

Suppose you want to use an Availability Group (AG). How is deploying a Windows Server Failover Cluster (WSFC) different? Are there any specific considerations for AGs that differ from an on-premises solution?

Implementing a PaaS-based high availability and disaster recovery (HADR) solution for PaaS solutions is different than IaaS. For IaaS, the configuration is done at the Azure level. For PaaS, the solutions that allow your applications and data to be highly available and meet your RTOs and RPOs are configured within the database or database server.

Implementing a PaaS-based HADR solution for PaaS solutions is different than IaaS. As noted above, for IaaS, the configuration is mainly done at the Azure level. For PaaS, the solutions that allow your applications and data to be highly available and meet your RTOs and RPOs are configured within the database or database server.

By the end of this module, you'll understand what is needed to be able to deploy IaaS database platform solutions in Azure.

Windows Server Failover Cluster

For AG and FCI, a cluster is required to protect against server or component failure (e.g., network failure, or RDBMS crashing)

For availability only, not for scale out performance

Enabled via the failover clustering feature

WSFCs managed via Failover Cluster Manager or PowerShell

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Deploying a WSFC in Azure is similar to configuring one on premises. There are some Azure-specific considerations which is what this section will focus on.

One of the most important aspects is deciding what to use for a witness resource. Witness is a core component of the quorum mechanism. Quorum is what helps ensure that everything in the WSFC stays up and running. If you lose quorum, the WSFC will go down taking an AG or FCI with it. The witness resource can be a disk, file share (SMB 2.0 or later), or cloud witness. It is recommended that you use a cloud witness as it is fully Azure-based, especially for solutions that will span multiple Azure regions or are hybrid. The cloud witness feature is available in Windows Server 2016 and later.

Considerations for deploying a WSFC in Azure

Witness:

A core component of quorum which helps ensure WSFC continues to run

Unless special requirements, one vNIC for each node in Azure

WSFC name and IP Address(es):

- WSFC requires a **unique name** different than any FCI, listener, or server in the environment
- One IP address per subnet for the WSFC itself
- WSFC name will require an **AD DS object** named the Cluster Name Object (CNO)
- WSFC name and IP address get an entry in DNS

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Witness recommendation: Cloud witness (Windows Server 2016 and later) to span Azure Regions or for hybrid

WSFC name and IP Address(es):

- WSFC requires a **unique name** different than any FCI, listener, or server in the environment
- One IP address per subnet for the WSFC itself
- WSFC name will require an **AD DS object** named the Cluster Name Object (CNO) for most configurations
- WSFC name and IP address get an entry in DNS

Unless special requirements, one vNIC for each node in Azure

It will show as a dynamic (DHCP) address and cluster validation will generate a warning that can safely be ignored

The next consideration is the Microsoft Distributed Transaction Coordinator (DTC or MSDTC). Some applications use it, but the majority do not. If you require DTC and are deploying an AG or FCI, you should cluster DTC. Clustering DTC requires a shared disk to work properly even though you may not require one otherwise, as in the case of an AG.

Most WSFC deployments require the use of both AD DS and DNS; FCIs always do. AGs can be configured without requiring AD DS but still needing DNS. In Windows Server 2016, there is a variant of a WSFC called a Workgroup Cluster, which can be used with AGs only. Additional considerations for this configuration will be discussed in the next section.

The WSFC itself needs a unique name in the domain (and DNS) and requires an object in AD DS called the cluster name object (CNO). Anything created in context of the WSFC that has a name will require a unique name, as well as at least one IP address. If the configuration will stay in a single region, IP addresses will be in a single subnet. If the WSFC will span multiple regions, more than one IP address will be associated with the WSFC, as well as an AG if it spans multiple regions as part of the WSFC.

Create a Failover Cluster: Checklist

Verify the prerequisites



Install the Failover Clustering feature on every server that you want to add as a cluster node



Run the Cluster Validation Wizard to validate the configuration



Run the Create Cluster Wizard to create the failover cluster



Create clustered roles to host cluster workloads

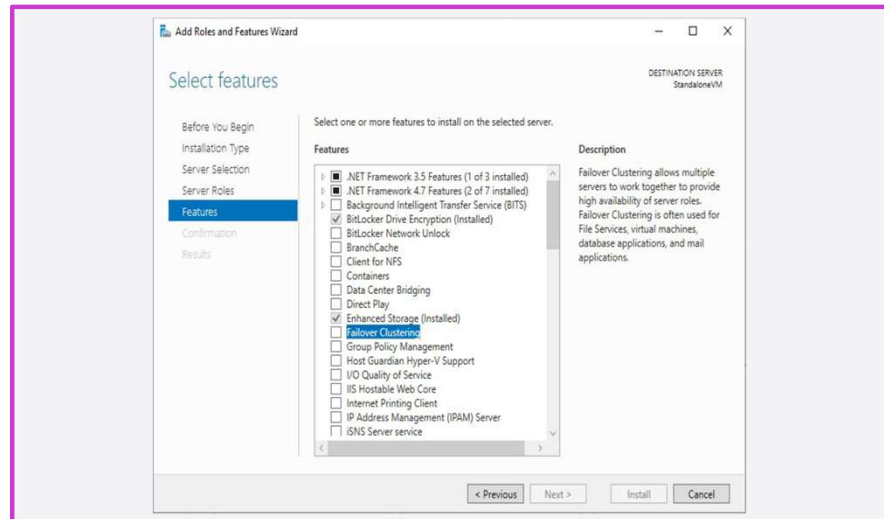
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Failover Clustering Feature

- Before a WSFC can be configured, the Failover Clustering feature must be enabled on every node that will use WSFC
- In Server Manager, Failover Clustering must be enabled in the Add Roles and Features Wizard

In PowerShell:

```
Install-WindowsFeature
Failover-Clustering -
IncludeManagementTools
```



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Cluster validation

Built-in process to test the configuration to ensure it is good

Required to create a WSFC

Must be run before clustering the VMs

Configuration must pass:

- **All green**
- **Warnings** only if not actually a problem or something unexpected
- **Failures** must be fixed
- **Not applicable** can be ignored

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For a WSFC to be considered supported, it must pass cluster validation. Cluster validation is a built-in process that checks the nodes via a series of tests to ensure that the entire configuration is suitable for clustering. After running validation, each of the tests will come back with an error, warning, pass, or not applicable. Warnings may be ok if that condition is expected in your environment. If not, it should be addressed. All errors must be resolved.

Validation can be run via Failover Cluster Manager or using the Test-Cluster PowerShell cmdlet.

For FCIs, these tests also check the shared storage to ensure that the disks are configured correctly. For AGs with no shared storage, in Windows Server 2016 and later, the results will come back as not applicable.

For more information on cluster validation, [see the official documentation](#).

Create and test an availability group

- The availability group creation process is the same as on premises, except for the process of creating the listener – Use SSMS, T-SQL, or PowerShell
- You may create the Azure Internal load balancer before or after creating the AG:
 - The ILB may be Basic or Standard – **Availability Zone deployments require Standard**
 - Create for the listener IP and the port used by SQL Server (not the AG endpoints)
 - Add the probe port to the IP address resource in the WSFC; will require a restart
- Test failover using SQL Server (T-SQL or SSMS)

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Once the listener is created, a load balancer must be used. Without configuring the load balancer, applications, end users, administrators, and more could not use the listener unless they were actually connected into the VM that is the primary replica for that AG. For full instructions on how to configure the load balancer, [see the official documentation](#).

You can use a basic or a standard load balancer depending on your preference or configuration. Note that deployments to Availability Zones require the use of a standard load balancer. . The listener IP address and the port used for the listener are what is configured as part of the load balancer. A single load balancer supports more than one IP address, so depending on your standards, you may not need a different load balancer for each AG configured on those nodes.

Considerations for availability groups in Azure

- Listener requires a **unique name** different from the WSFC, nodes, or anything else, as well as an **IP per subnet**
 - Cannot reserve the listener's IP address in Azure
- Deploying SQL Server VMs to multiple subnets is considered a best practice to avoid reliance on Azure Load Balancer or DNN for HADR traffic routing
- AGs configurations require an underlying cluster, like an on-prem setup
- All VMs in AG should have the same storage configuration

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[Tutorial: Configure a SQL Server Always On availability group - SQL Server on Azure VM | Microsoft Docs](https://learn.microsoft.com/azure/azure-sql/virtual-machines/windows/availability-group-manually-configure-tutorial):
https://learn.microsoft.com/azure/azure-sql/virtual-machines/windows/availability-group-manually-configure-tutorial

[Configure availability group \(Azure quickstart template\) - SQL Server on Azure VM | Microsoft Docs](https://learn.microsoft.com/azure/azure-sql/virtual-machines/windows/availability-group-quickstart-template-configure):
https://learn.microsoft.com/azure/azure-sql/virtual-machines/windows/availability-group-quickstart-template-configure

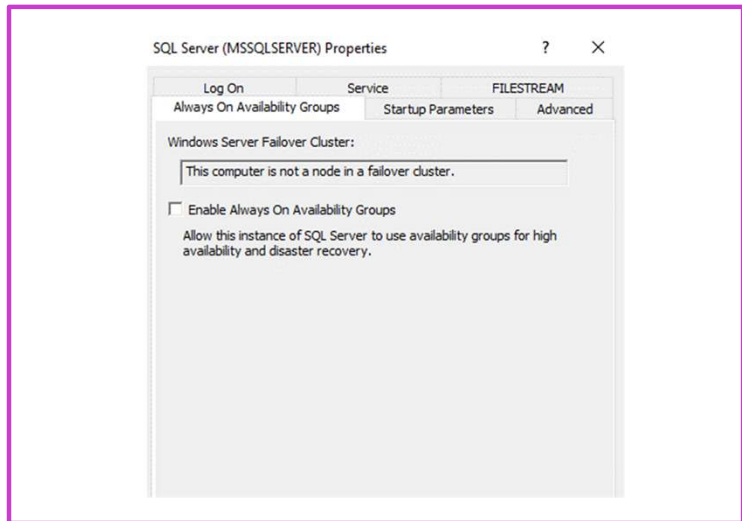
[Tutorial: Configure a SQL Server Always On availability group - SQL Server on Azure VM | Microsoft Docs](https://learn.microsoft.com/azure/azure-sql/virtual-machines/windows/availability-group-manually-configure-tutorial): <https://learn.microsoft.com/azure/azure-sql/virtual-machines/windows/availability-group-manually-configure-tutorial>

Configuring an AG is nearly the same in Azure as it is on premises as are most of the considerations, such as how to initialize secondary replicas. Most of the Azure-specific considerations are covered in the WSFC section, and specifically names and IP addresses. If using a static IP address for the listener, there is no way to reserve it in Azure so you will need to monitor for conflicts. The name for the listener must be unique and different from the name of the WSFC. Each AG will have one listener.

As mentioned earlier, do not place any permanent database on the ephemeral storage. All VMs that are participating in an AG should have the same storage configuration. You must size disks appropriately for performance depending on the application workload.

Enable the availability groups feature

- Cannot configure an AG without enabling the feature in **SQL Server Configuration Manager** or with `Enable-SqlAlwaysOn` in PowerShell
- **Must** be done after configuring the WSFC
- Requires instance **restart**



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Before an AG can be configured, similar to a WSFC, the feature must be enabled. This can be done in SQL Server Configuration Manager as shown in the dialog below or via PowerShell with the cmdlet [Enable-SqlAlwaysOn](#). Enabling the AG feature will require a stop and start of the SQL Server service.

Distributed Availability Groups

- ✓ Like a regular AG, same creation process as on-premises
- ✓ Requires creation of multiple WSFCs, AGs, and listeners
- ✓ Requires an ILB for each WSFC/AG pair

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Distributed Availability Groups är en utökning av Always On Availability Groups som gör att du kan koppla ihop flera separata AG-miljöer, ofta mellan datacenter eller Azure och on-premises. De fungerar i grunden som en vanlig AG, och skapandet liknar därför den etablerade processen man redan kan från lokala installationer. Skillnaden är att lösningen kräver flera Windows Server Failover Clusters (WSFC), flera AG:er samt listeners, vilket gör arkitekturen mer komplex. I Azure behövs dessutom en ILB (Internal Load Balancer) för varje WSFC/AG-par för att hantera trafiken till respektive listener. Resultatet blir en robust lösning för DR och failback mellan platser.

Planning for and configuring a distributed AG is the same on premises as it is in Azure, with any Azure-specific considerations for the individual AGs. The main difference between on premises and Azure for a distributed AG is that as part of the load balancer configuration in each region, the endpoint port for the AG needs to be added. The default port is 5022.

HADR options for PaaS offering



Active geo-replication



Failover groups

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Active geo-replication is configured per database. To fail over a group of databases, or if your application requires a stable connection endpoint, consider [Failover groups](#) instead.

The failover groups feature allows you to manage the replication and failover of some or all databases on a [logical server](#) to a logical server in another region. This article provides an overview of the failover group feature with best practices and recommendations for using it with Azure SQL Database.

Active geo-replication



Programmatically or manually failover primary databases to secondary regions during major disaster



Primary and secondary replicas are required to have the same **service tier and compute size**



Cross subscription replication supported – Configure a secondary replica on a different subscription than the primary database



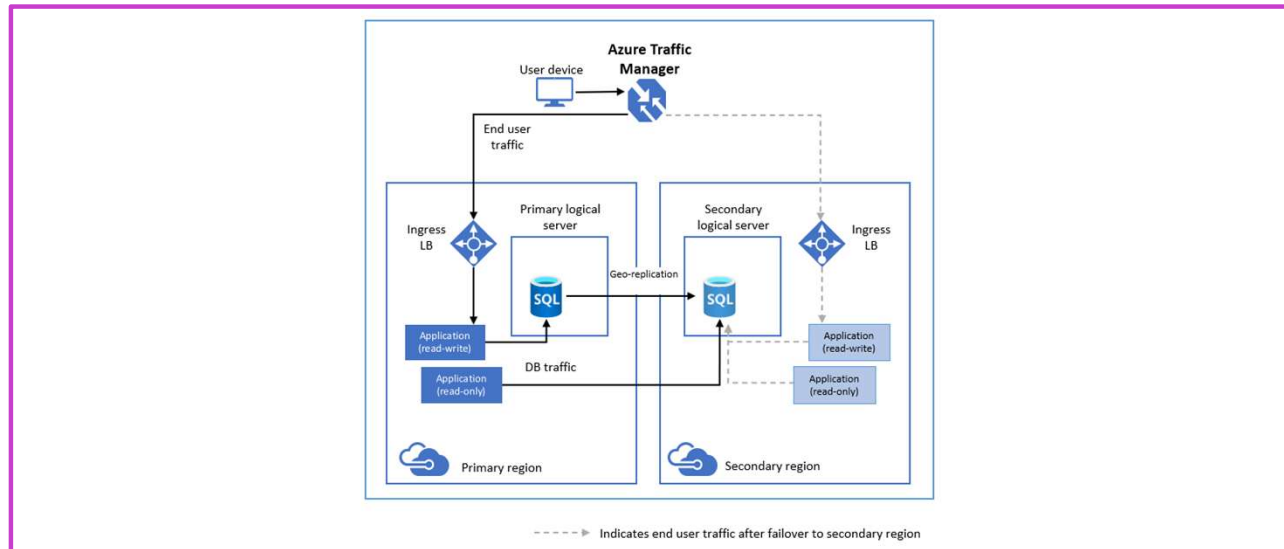
Only supported on Azure SQL Database

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One method to increase availability for Azure SQL Database is to use active geo-replication. Active geo-replication creates a secondary database replica in another region that is asynchronously kept up to date.

This replica is readable, similar to an Availability Group in IaaS. Underneath the surface, Azure uses Availability Groups to maintain this functionality, which is why some of the terminologies are similar (primary and secondary logical servers, read-only databases, etc.).

Active Geo-Replication cont'd



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If your application requires a stable connection endpoint and automatic geo-failover support in addition to geo-replication, use Auto-failover groups.

If for any reason your primary database fails, you can initiate a geo-failover to any of your secondary databases. When a secondary is promoted to the primary role, all other secondaries are automatically linked to the new primary.

You can manage geo-replication and initiate a geo-failover using the following:

- The [Azure portal](#)
- [PowerShell: Single database](#)
- [PowerShell: Elastic pool](#)
- [Transact-SQL: Single database or elastic pool](#)
- [REST API: Single database](#)

Active Geo-Replication: Manual failover

[+ Create replica](#) [Refresh](#) [Feedback](#)

Geo replicas for your database are listed below. Geo replicas reside on a different logical server from the primary and protect against regional failures or prolonged data center outage. [Learn more](#)

Name ↑↓	Server ↑↓	Region ↑↓	Failover policy ↑↓	Pricing tier ↑↓	Replica state ↑↓	
▼ Primary						
AdventureWorksLT	dp300-lab-23075881	South Central US	None	General Purpose: Gen5, 2 vCores	Online	...
▼ Geo replicas						
AdventureWorksLT	dp300-lab-dr-23075881	East US		General Purpose: Gen5, 2 vCores	Readable	...
						Pin to dashboard ☐ Stop replication ☒ Forced failover

After the secondary replica is created, you can manually fail over your secondary replica. The roles will switch with the secondary becoming the new primary, and the old primary the secondary.

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Auto Failover Groups



Database is created automatically on the secondary through a process called **seeding**



It can contain one or more databases



Depending on the size of the database, the seeding process may take some time



If you have a tight RPO and can't afford much data loss, set the **GracePeriodWithDataLossHours** property to a higher value (default is 1-hour)



Supported on SQL Database and SQL Managed Instance

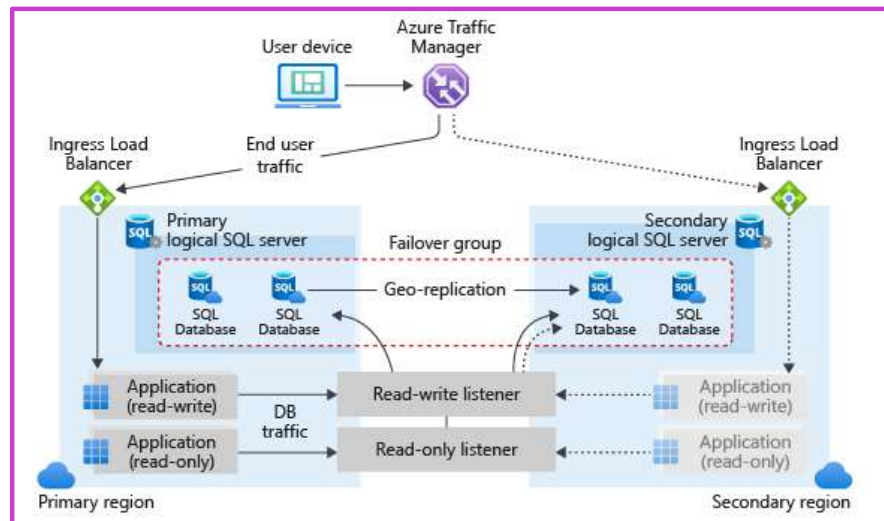
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An auto-failover group is an availability feature that can be used with both Azure SQL Database and Azure SQL Managed Instance. Autofailover groups let you manage how databases on an Azure SQL Database server or databases in SQL Managed Instance are replicated to another region, and let you manage how failover could happen. The name assigned to the autofailover group must be unique within the *.database.windows.net domain. SQL Managed Instance only supports one autofailover group.

Auto-failover groups provide AG-like functionality called a listener, which allows both read-write and read-only activity. This functionality can be seen in the image below which is slightly different than the one for active geo-replication. There are two different kinds of listeners: one for read-write and one for read-only traffic. Behind the scenes in a failover, DNS is updated so clients will be able to point to the abstracted listener name and not need to know anything else. The database server containing the read-write copies is the primary, and the server that is receiving the transactions from the primary is a secondary.

Auto Failover Groups – Azure SQL Database

- There are two different kinds of listeners:
 - For **read-write**
 - For **read-only** traffic
- The secondary's database is created automatically through **seeding**
- Behind the scenes in a failover, DNS is updated so clients will be able to point to the abstracted listener name



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The auto-failover group must be configured on the primary server and will connect it to the secondary server in a different Azure region. The groups can include all or some databases in these servers. The following diagram illustrates a typical configuration of a geo-redundant cloud application using multiple databases and auto-failover group.

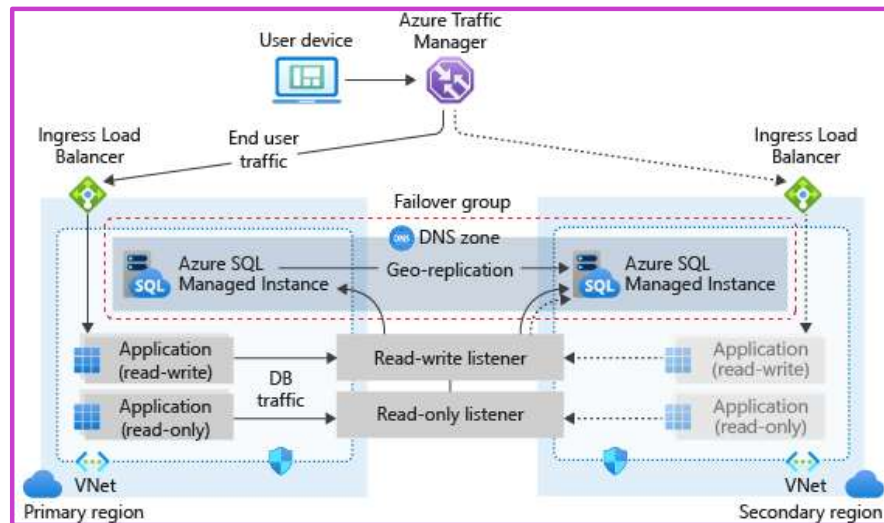
When configuring a failover group, ensure that authentication and network access on the secondary is set up to function correctly after geo-failover, when the geo-secondary becomes the new primary.

Auto-failover groups have two different policies that can be configured.

- **Automatic** – By default, when a failure occurs and it's determined that a failover must happen, the auto-failover group will switch regions. The ability to fail over automatically can be disabled.
- **Read-Only** – By default, if a failover occurs, the read-only listener is disabled to ensure performance of the new primary when the secondary is down. This behavior can be changed so that both types of traffic are enabled after a failover.

Auto Failover Groups – Azure SQL Managed Instance

- All user databases in the instance will be replicated to the secondary instance
- System databases like *master* and *msdb* will **not** be replicated
- Both the primary and secondary instances must be in the same DNS zone
- SQL Managed Instance failover groups in paired regions have better performance compared to unpaired regions
- Number of databases involved may affect seeding duration



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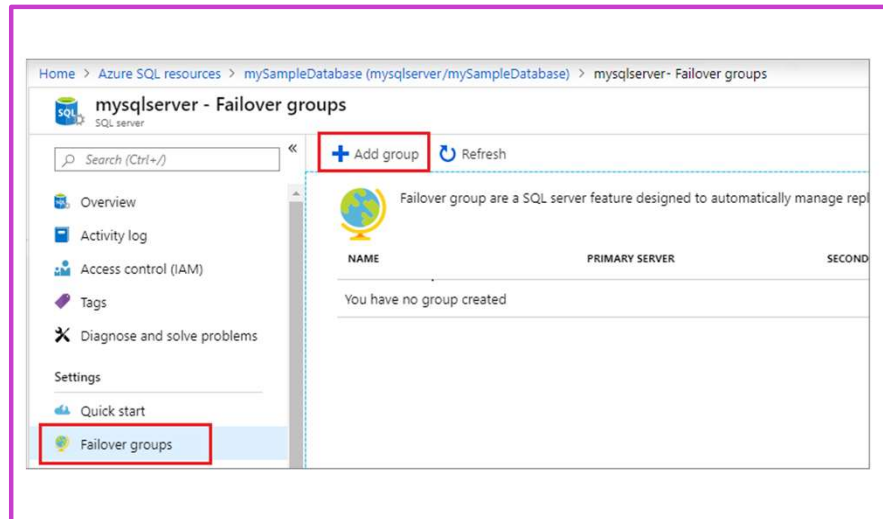
The auto-failover group must be configured on the primary instance and will connect it to the secondary instance in a different Azure region. All user databases in the instance will be replicated to the secondary instance. System databases like *master* and *msdb* will not be replicated.

The following diagram illustrates a typical configuration of a geo-redundant cloud application using managed instance and auto-failover group

Configure an auto-failover group for Azure SQL Database

Step 1:

Select **Failover groups** under the **Settings** pane, and then select **Add group** to create a new failover group



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The auto-failover group must be configured on the primary instance and will connect it to the secondary instance in a different Azure region. All user databases in the instance will be replicated to the secondary instance. System databases like *master* and *msdb* will not be replicated.

The following diagram illustrates a typical configuration of a geo-redundant cloud application using managed instance and auto-failover group

Configure an auto-failover group cont'd

Step 2:

On the **Failover Group** page, enter or select the required values, and then select **Create**

Adding the database to the failover group will automatically start the geo-replication process

The screenshot shows the 'Failover group' configuration page in the Azure portal. The 'Database within the group' section is highlighted with a red box, showing 'mySampleDatabase' as the selected database. The 'Databases' panel on the right shows a table with columns: NAME, ROLE, SECONDARY SERVER, and STATUS. The table contains one row for 'mySampleDatabase' with status 'Online'. Below the table, there is a 'Summary' section showing 'Databases on secondary (excluding ones in Elastic Pools)' and 'Elastic Pools on secondary server' both with a count of 0. A 'Monthly cost' section is also visible.

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The auto-failover group must be configured on the primary instance and will connect it to the secondary instance in a different Azure region. All user databases in the instance will be replicated to the secondary instance. System databases like *master* and *msdb* will not be replicated.

The following diagram illustrates a typical configuration of a geo-redundant cloud application using managed instance and auto-failover group

Auto Failover Groups vs. Geo-Replication

	Geo-replication	Failover groups
Automatic failover	No	Yes
Fail over multiple databases simultaneously	No	Yes
User must update connection string after failover	Yes	No
SQL Managed Instance support	No	Yes
Can be in same region as the primary	Yes	No
Support multiple replicas	Yes	No
Support read-scale	Yes	Yes

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Knowledge check



What component in Azure needs to be configured for the listener in an AG to work properly?

- ☐ The NIC
- ☐ The NSG
- ☒ A load balancer



What feature creates a replica of the database in another region that is asynchronously kept up to date?

- ☐ Failover group
- ☐ Readable secondary replica
- ☒ Active geo-replication



What setting for auto-failover groups must be changed to ensure a low RPO?

- ☐ RPOZero
- ☐ AlwaysBeInSync
- ☒ GracePeriodWithDataLossHours

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1) What component in Azure needs to be configured for the listener in an AG to work properly?



Correct: A load balancer

Förklarar listener-IP och dirigerar trafik till aktiv nod.



Incorrect: The NIC

NIC ger nätverk, men skapar inte AG-listenerns virtuella IP.



Incorrect: The NSG

NSG styr trafikregler, men ersätter inte load balancer-funktionen.

2) What feature creates a replica of the database in another region that is asynchronously kept up to date?



Correct: Active geo-replication

Skapar sekundär databas i annan region, asynkront uppdaterad.



Incorrect: Failover group

Failover group använder geo-replication, men är inte själva repliken.



Incorrect: Readable secondary replica

Gäller SQL AG sekundärnod, inte Azure SQL geo-replikering.

3) What setting for auto-failover groups must be changed to ensure a low RPO?



Correct: GracePeriodWithDataLossHours

Sänks för att minska tillåten dataförlust vid failover.



Incorrect: RPOZero

Finns ingen sådan inställning i auto-failover groups.



Incorrect: AlwaysBeInSync

Låter bra men existerar inte som faktisk Azure-inställning.

Instructor led labs: Configure geo-replication for Azure SQL Database

Enable geo-replication
Failover to a secondary region

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<https://microsoftlearning.github.io/dp-300-database-administrator/Instructions/Labs/14-configure-geo-replication-for-azure-sql-database.html>

Back up and restore databases

<https://learn.microsoft.com/training/modules/backup-restore-databases/>

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Even if you do nothing else to increase your availability with features like AGs or active geo-replication, you must have a solid backup and recovery strategy. If for some reason a feature fails, you will have to restore databases from backups.

In this lesson, you will learn the various options on how to implement database backup and restore in Azure.

Objectives



Backup and restore options for IaaS



Backup and restore options for PaaS

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Even if you do nothing else to increase your availability with features like AGs or active geo-replication, you must have a solid backup and recovery strategy. If for some reason a feature fails, you will have to restore databases from backups.

Backup and restore fundamentals for IaaS

SQL Server backup types:

Full – Backs up the entire whole database, but does not flush the transaction log

Differential – All changes since last full backup

Transaction log – Backs up the transactions in the log since the transaction log backup

- Database must be in full or bulk-logged recovery model to generate a transaction log backup

Do not use SIMPLE recovery model if you require point in time recovery

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SQL Server databases have the concept of a recovery model which governs the type of backups and restores that could be achieved for a database. The recovery models are Full, Bulk-logged, and Simple. Most databases are set to Full or Simple. Full allows all types of backups to be generated while Simple does not allow transaction log backups. This means that if you require point in time recovery, Simple may not meet your needs as you cannot restore to a granular point in time.

Full Azure Virtual Machine backup

- Backs up full VM including SQL Server databases via VSS
Restores VM as a whole
- Creates an application-consistent backup which will not 'break' SQL Server
Combining SQL Server backups with snapshots can potentially cause issues
- May cause problems for some databases, always test the strategy
- Ensure it can meet RTO and RPO
- Causes a freeze/thaw of I/O of SQL Server log

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[Create an Azure snapshot of a virtual hard disk - Azure Virtual Machines | Microsoft Docs](https://learn.microsoft.com/azure/virtual-machines/snapshot-copy-managed-disk?tabs=portal)
<https://learn.microsoft.com/azure/virtual-machines/snapshot-copy-managed-disk?tabs=portal>

Azure Backup can back up VMs that contain SQL Server. These backups would not be just SQL Server databases; it would be everything that is in the VM so it could be restored as a whole. While this option may not be right for everyone, it can potentially protect against things like ransomware.

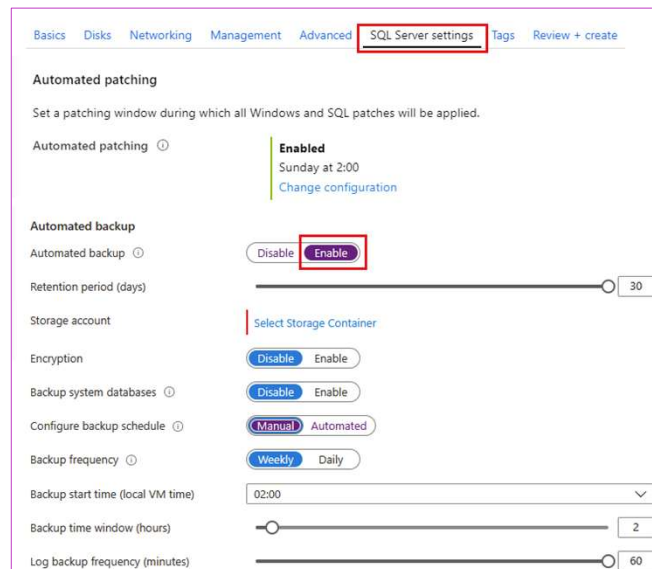
These types of backups are SQL Server aware via VSS so it will create a crash-consistent backup that will not break SQL Server if restored. If using this option, when looking in the SQL Server log you will see that the I/O has been momentarily frozen and then starts again when complete. If this causes issues with availability features like AGs, you may want to consider another backup strategy.

Combining SQL Server backups with snapshots can potentially cause issues. If snapshot delays cause backup failures, set following registry key:

```
[HKEY_LOCAL_MACHINE\SOFTWARE\MICROSOFT\BCDRAGENT]
"USEVSSCOPYBACKUP"="TRUE"
```

Automated backups using SQL Server resource provider

- Uses an Azure storage account as a target
- Azure manages the SQL Server backups (SQL VM)-
- Any IaaS VM that has SQL Server installed can use the **SQL Server resource provider**
- Can configure schedule, frequency, and retention
- **Choose one method to back up databases**



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Ja – **exakt** 👍

Det är **backup av SQL Server-databaser som körs i en VM** (SQL Server on Azure VM / IaaS).

Alltså:

Källan: databaserna inne i SQL Server på VM:n

Målet: en **Azure Storage Account / Blob container**

Det som backas upp: **.bak (full), diff, och log-backuper**

Inte: hela VM:n (disk-snapshots)

Så man kan säga: **"databasbackup från en VM"**, inte **"VM-backup"**.

- Only available for Windows Server-based SQL Server installations.
- **Choose one method to back up databases** – i.e. it could cause problems if you have additional backup configured inside the VM as the log chain is cleared out when it is backed up.

Back up to Local disk or network share

- Just like on premises; can perform native backups to disks attached to the VM or to a network share
- Network share can be on Azure-based storage
- If local disk, still need to ensure backups are periodically copied off of the VM for redundancy

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Backup to and restore from URL

- Backup SQL Server databases directly to or restore from Azure blob storage
- Backup to URL can be used by **on premises** as well as IaaS instances
- Backup uses geo-replicated (GRS) storage accounts for another layer of data protection
- Requires authentication between the instance and Azure Storage
- SSMS, T-SQL, and PowerShell support

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- Backup to URL can be used by **on premises** as well as IaaS instances. Requires an Azure storage account and an Azure blob container.
- Requires authentication between the instance and Azure Storage. Configure it with a SQL Server Credential:
Recommended credential: Shared Access Signature offers better security control and uses block blobs (less expensive).
 Will also work with an Azure storage account and access key authentication and uses **page blob**.
- SSMS, T-SQL, and PowerShell support. T-SQL example:
`BACKUP LOG contoso`
`TO URL = 'https://myacc.blob.core.windows.net/mycontainer/contoso202003271200.trn'`

Another option is to configure backup to URL for the SQL Server instance installed in the VM. Unlike on premises, backup and restore from URL for an IaaS VM is effectively a local option.

Backup to URL requires an Azure storage account and uses the Azure blob storage service. Inside the storage account there are containers, and the blobs are stored there. Unlike a path on your local disk, the path to a backup file will look something like <https://ACCOUNTNAME.blob.core.windows.net/ContainerName/MyDatabase.bak>. You can include additional folder names under your container for easier identification of backups (e.g. FULL, DIFF, LOG). These folders are virtual and are properties of the file

Database backup and restore for Azure SQL Database

- Backups automatically generated: Full once a week, differential ever 12-24 hours, t-logs every 5-10 min
- Stored in **RA-GRS** blobs that are replicated to another datacenter
- Configured per database
- No user driven backup option
- Restore is per database
- It is possible to restore a deleted database

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Configure policies

SQL server

Point-in-time-restore

Specify how long you want to keep your point-in-time backups. [Learn more](#)

How many days would you like PITR backups to be kept?

7

Differential backup frequency

Specify how often you want differential backups to be taken. [Learn more](#)

Take a differential backup every:

12 Hours

Long-term retention

Specify how long you want to keep your long-term retention backups. You may choose to keep yearly backups for up to 10 years. [Learn more](#)

Weekly LTR Backups

Keep weekly backups for:

0

Week(s)

Monthly LTR Backups

Keep the first backup of each month for:

0

Week(s)

Yearly LTR Backups

Keep an annual backup for:

0

Week(s)

Which weekly backup of the year would you like to keep?

Week 1

Apply

Cancel

Database backups are an essential part of any business continuity and disaster recovery strategy because they help protect your data from corruption or deletion.

SQL Database can assist you to be compliant with mandatory backups for regulatory purposes with retention policies.

Point in time restore

- Restore databases to a specific point in time to recover from errors, failures, or data loss
- Point in time retention policy is set to seven days, and you can change it to up to 35-days.
- Cannot restore an entire SQL Managed Instance deployment; only individual databases.

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Home > SQL databases > AdventureWorks (contoso/AdventureWorks) > contoso > AdventureWorks (contoso/AdventureWorks)

Create SQL Database - Restore database

Microsoft

Basics Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription

Resource group

Source Details

Select a backup source and details. Additional settings will be defaulted where possible based on the backup selected.

Source Database

Select source

Earliest restore point

Restore point (UTC) *

Choose a restore point between the earliest restore point and the current time in UTC.

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources.

Database name *

Server

Want to use SQL elastic pool? * ☐ Yes ☒ No

Compute + storage *

Serverless, Gen5, 2 vCores, 32 GB storage

[Configure database](#)

For Azure SQL Database:

- You can't overwrite an existing database during restore.
- Database restore operations don't restore the tags of the original database.

Database backup and restore for Azure SQL Managed Instance



Database backups automatically generated by default



Can restore to a point in time; like Azure SQL Database



Manually generate backups via backup to URL:

Requires credentials to access Azure Blob Storage

Can Restore too

Full backups can only generate a COPY_ONLY backup to maintain the log chain, i.e.

`BACKUP DATABASE contoso`

`TO URL = 'https://myacc.blob.core.windows.net/testcontainer/contoso.bak' WITH COPY_ONLY`



Backups generated cannot be used with Azure SQL Database

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By default, Azure creates backups for databases in Azure SQL Database Managed Instance and functions like Azure SQL Database. To be able to restore a database to a previous point in time, follow the [official documentation](#).

You can also manually back up and restore databases with Azure SQL Database Managed Instance using the same backup to URL/restore from URL functionality found in SQL Server covered earlier. That requires the use of credentials to access Azure Blob Storage.

There is one major difference: for full backups, you can only generate a COPY_ONLY backup since Azure SQL Database Managed Instance is maintaining the log chain. A sample backup statement would look like

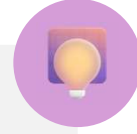
`BACKUP DATABASE contoso`

`TO URL = 'https://myacc.blob.core.windows.net/testcontainer/contoso.bak'`

`WITH COPY_ONLY`

More information on backing up databases using backup to URL with Azure SQL Database Managed Instance can be found [here](#), and restoring [here](#).

Knowledge check



What is the destination of a backup to URL in SQL Server or Azure SQL Server Database Managed Instance?

- ☐ URL
- ☒ Blob
- ☐ Pointer



What is the default redundancy option for Azure SQL databases?

- ☐ Locally-redundant backup storage (LRS)
- ☐ Zone-redundant backup storage (ZRS)
- ☒ Geo-redundant backup storage



Which option should you use to geographic failover instances of SQL Managed Instance?

- ☒ Auto-failover groups
- ☐ Active geo-replication
- ☐ Failover Cluster Instance

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1) What is the destination of a backup to URL in SQL Server or Azure SQL Server Database Managed Instance?

☒ **Correct: Blob**

Backup sparas som fil i Azure Blob Storage-container.

☒ **Incorrect: URL**

URL är adressen, inte själva lagringsplatsen för backupen.

☒ **Incorrect: Pointer**

Ingen "pekare"; backupen lagras fysiskt som blob-objekt.

2) What is the default redundancy option for Azure SQL databases?

☒ **Correct: Geo-redundant backup storage**

Standard är geo-redundanta backuper i parregion för skydd.

☒ **Incorrect: Locally-redundant backup storage (LRS)**

LRS är möjligt val, men inte standard för Azure SQL.

☒ **Incorrect: Zone-redundant backup storage (ZRS)**

ZRS är valbart i vissa regioner, men ej default.

3) Which option should you use to geographic failover instances of SQL Managed Instance?

☒ **Correct: Auto-failover groups**

Ger geo-failover mellan Managed Instances i olika regioner.

☒ **Incorrect: Active geo-replication**

Gäller Azure SQL Database, inte SQL Managed Instance.

☒ **Incorrect: Failover Cluster Instance**

FCI är lokal HA inom region, inte geografisk failover.

Instructor led labs: Backup to URL and restore from URL

Create a credential
Backup database to URL
Validate backup through Azure CLI
Restore from URL

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References

Business continuity and HADR for SQL Server on Azure Virtual Machines:

<https://learn.microsoft.com/azure/azure-sql/virtual-machines/windows/business-continuity-high-availability-disaster-recovery-hadr-overview>

Restore a database from a backup in Azure SQL Database:

<https://learn.microsoft.com/azure/azure-sql/database/recovery-using-backups>

Monitor backup activity for Azure SQL Managed Instance:

<https://learn.microsoft.com/azure/azure-sql/managed-instance/backup-activity-monitor>

Configure a failover group for Azure SQL Database:

<https://learn.microsoft.com/azure/azure-sql/database/failover-group-configure-sql-db>

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Learning Path Recap

In this learning path, we:

Learned how to describe high availability and disaster recovery strategies. This involves understanding Recovery time objective (RTO) and Recovery point objective (RPO), understanding the available HADR options for both IaaS and PaaS, and building a HADR strategy.

We also learned how to back up and restore databases. This includes exploring backup and restore options for IaaS, as well as backup and restore options for PaaS.

Lastly, we explored how to deploy IaaS and PaaS solutions for high availability and disaster recovery. This involves understanding the considerations for deploying a Windows Server Failover Cluster in Azure, the considerations for deploying an Availability Group in Azure, Temporal Tables, Active-geo replication, and Auto-failover groups.

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Please complete the training survey below!

DP-300: Administering Microsoft Azure SQL Solutions

Please insert training survey link as appropriate

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Note:

- For repeatability, a trainer may also want to use *aka.ms* and then change the survey link before each delivery. For example, *aka.ms/az-900/survey*, etc.
- A Learning Partner/MCT should input an MTM survey link or other survey link as appropriate



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